Learning stochastic differential equations using RNN with log signature features

Based on paper by Shujian Liao, Terry Lyons, Weixin Yang, Hao Ni

Konstantins Starovoitovs (HU Berlin)

June 2021

Outline

Theoretical setup

Setup

Continuous RNN has the form (for τ is a constant, A and B are matrices and σ is a non-linear activation function)

$$\dot{Y}_t = -\frac{Y_t}{\tau} + A\sigma\left(BY_t\right) + I_t.$$

We can rewrite it as an **RDE** (with $X_t = \left(t, \int_{c=t_0}^{t} I_s ds\right)$ and $V(y,(t,x)) = -\frac{y}{\tau}t + A\sigma(By)t + x)$

$$dY_t = V\left(Y_t\right)dX_t.$$

Objective: model a functional on the above stream.

Log-signature

Consider element in the tensor algebra

$$X_J^I = \int_{\substack{u_1 < \cdots < u_k \\ u_1, \ldots, u_k \in J}} dX_{u_1}^{(i_1)} \otimes \cdots \otimes dX_{u_n}^{(i_n)}$$

Define **signature** as

$$S(X)_J = \left(1, \mathrm{X}_J^1, \ldots, \mathrm{X}_J^k, \ldots\right)$$

Log-signature is the induced via logarithmic map

$$\log(a) = \log(1 + t) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} t^{\otimes n} \qquad \forall a \in T(E).$$

Features of the log-signature

- Representation of complex un-parameterized streams of multi-modal data over a coarse time scale with a low dimensional representation.
- **Bijection** between signature and log-signature (via exponential map), while thanks to algebra log-signature can be represented via commutators in a **parsimonious** way (e.g. $[e_i, e_j] = -[e_j, e_i]$). *For example, for* $d = 102$. *Signature dimension* $= 102 + 102^2 = 10506$. *Log-signature gets away with half that amount.*
- Invariance under time reparametrization.
- Characterizes stream up to negligible equivalence class.

	signature	log-signature
uniqueness of a path	✓	✓
invariance of time parameterization	✓	✓
the universality	✓	✗
no redundancy	✗	✓

Figure: Comparison of signature and log-signature

Model architecture

Logsig-RNN illustration

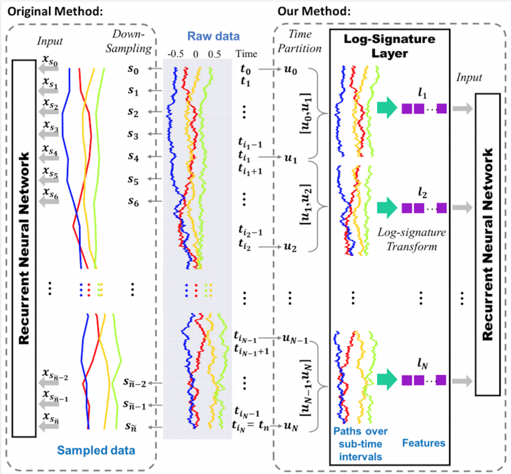

Figure: Comparison of Logsig-RNN and RNN

Ingredients

- Segmented **log-signature layer** with backpropagation (thus can be inserted in between the layers, not just feature extraction)
- Connect log-signature input with RNN architecture while retaining universality property **(Logsig-RNN)**
- Embed high-dimensional signal into smaller space with a series of path transformations **(PT-Logsig-RNN)**

Logsig-RNN features

- **Reduction of the time dimension** High-frequency sample transformed into coarser time partition.
- **High frequency and continuous data** If RNN is applied to the time series directly, the signal has to be downsampled, so that microscopic characteristics of the data might get lost.
- **Robustness to missing data** Compared to the signature feature, log-signature is a parsimonious representation of the signature.
- **Highly oscillatory stream data** Two quite different highly oscillatory data streams can look nearly identical once sampled at very fine levels, thus for the vanilla RNN approach to work one would require lots of augmentation and sampling at very fine levels

PT-Logsig-RNN

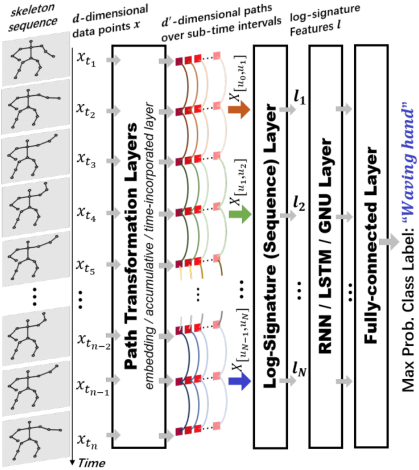

Figure: PT-Logsig-RNN architecture

Log-signature layer

Consider a fine partition $\hat{\mathcal{D}}$ and a coarser sub-partition $\mathcal{D}$. Log-signature layer is a mapping

$$L : \mathbb{R}^{d \times n} \to \mathbb{R}^{d_l \times N} = \left(lS_M(x)_{[u_k, u_{k+1}]} \right)_{k \in \{0 \cdots N-1\}}$$

where d_l is the dimension of the corersponding log-signature.

Remark. The log-signature layer does not have any weights.

Backpropagation

Consider functional F on the log-signature. By chain rule, we have

$$\frac{\partial F\left(\left(l_1, \cdots, l_N\right)\right)}{\partial x_{t_i}} = \sum_{k=1}^{N} \frac{\partial F\left(l_1, \cdots, l_N\right)}{\partial l_k} \frac{\partial l_k}{\partial x_{t_i}}.$$

If $t_i \notin [u_{k-1}, u_k]$, then $\frac{\partial l_k}{\partial x_{t_i}} = 0$. Otherwise, the log-signature is differentiable w.r.t. x_{t_i} and the algorithm is given in x. Thus

$$\frac{\partial l_k}{\partial x_{t_i}} = 1_{t_i \in \left[u_{k-1}, u_k\right]} \nabla_{x_{t_i}} LS\left(x_{u_{k-1}, u_k}\right).$$

Recurrent unit

- RNNs are used to model temporal sequences.
- Mathematically, simplest recurrent unit RNN can be written as a sequence of the hidden and output states

$$h_t = \sigma\left(Ux_t + Wh_{t-1}\right) \qquad o_t = q\left(Vh_t\right)$$

- One fits kernels $\Theta := \{U, W, V\}$ (which do not depend on time) and biases in the activation functions.
- Ubiquitous choice for activations: tanh and sigmoid due to GPU implementation.
- Alternatively, one can use other recurrent units (RNN/LSTM/GRU).

Different recurrent units

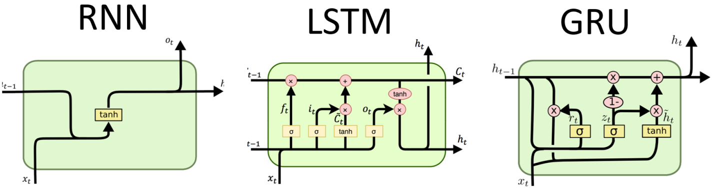

Figure: RNN vs. LSTM vs. GRU

[1] Image courtesy: https://stats.stackexchange.com/a/420172/193999

Path transformation layer

For a 102-dimensional path, degree of the 2nd order signature is 10506 and 2nd order degree of the log-signature is 5253.

- **Linear embedding layer.** Transform the input (X_{t_i}) to (LX_{t_i}) for a linear map L with range of lower dimension.
- **Accumulative layer.** Map sequence to its partial sum $Y_{t_i} = \sum_{j=1}^{i} X_{t_j}$ in order to capture quadratic variation and higher order statistics.
- **Time-incorporated layer.** Add the time dimension to the input sequence mapping (X_{t_i}) to (t_i, X_{t_i}). The log-signature of the time-incorporated transformation of a path is proved to fully recover the path.

Implementation

Implementation details

- Add starting points of the segment.
- Batch normalization after the Logsig layer.
- Dropout after the LSTM layer.

Human action recognition model

Layer	Output shape	Discription
Input	$(72, 25, 6)$	
Conv2D	$(72, 25, 32)$	Kernel size=$1 \times 1 \times 32$
Conv2D	$(68, 25, 16)$	Kernel size=$5 \times 1 \times 16$
Conv1D	$(68, 40)$	Kernel size=$1 \times 400 \times 40$
Logsig	$(32, 820)$	$M = 2, N = 4$
Add Starting Points	$(32, 860)$	Starting points of Conv1D output
LSTM	$(32, 256)$	Return sequential output
Output	120	

Figure: Architecture of the human action recognition model.

Each convolutional layer has linear activations.

Degree of the log-signature

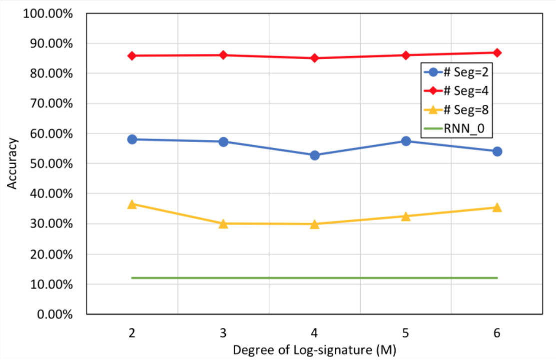

Figure: Validation of the degree of the log-signature

NB: corresponds to a different problem / dataset.

ICCV 2021

Model architecture in Tensorflow

Layer (type)	Output Shape	Param #	Connected to
input_1 (InputLayer)	[(None, 305, 17, 4)]	0	
conv2d (Conv2D)	(None, 305, 17, 32)	160	input_1[0][0]
conv2d_1 (Conv2D)	(None, 301, 17, 16)	2576	conv2d[0][0]
reshape (Reshape)	(None, 301, 272)	0	conv2d_1[0][0]
conv1d (Conv1D)	(None, 301, 40)	10920	reshape[0][0]
lambda_1 (Lambda)	(None, 32, 820)	0	conv1d[0][0]
reshape_1 (Reshape)	(None, 32, 820)	0	lambda_1[0][0]
lambda (Lambda)	(None, 32, 40)	0	conv1d[0][0]
batch_normalization (BatchNorma	(None, 32, 820)	3280	reshape_1[0][0]
concatenate (Concatenate)	(None, 32, 860)	0	lambda[0][0] batch_normalization[0][0]
lstm (LSTM)	(None, 32, 64)	236800	concatenate[0][0]
dropout (Dropout)	(None, 32, 64)	0	lstm[0][0]
flatten (Flatten)	(None, 2048)	0	dropout[0][0]
dense (Dense)	(None, 155)	317595	flatten[0][0]

Total params: 571,331
Trainable params: 569,691
Non-trainable params: 1,640

Figure: Model architecture in Tensorflow

Loss and accuracy

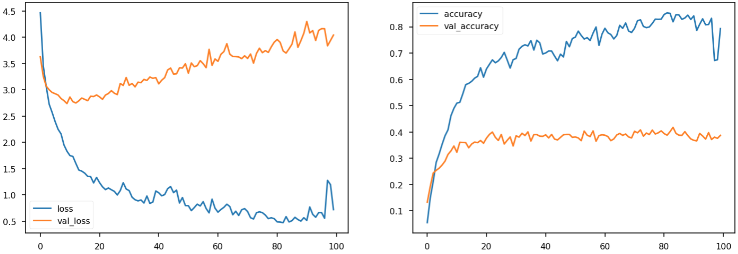

Figure: Learning curves.

Samples of different size

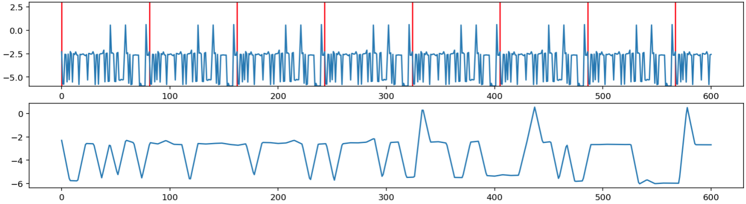

Figure: Samples of smaller size are repeated.

Bad for log-signature?

Dyadic log-signatures

Incorporate simultaneously log-signatures for different segment sizes.

- For each segment size (4, 8, 16, 32, 64, ...) consider an RNN unit (in parallel circuit), concatenate output and feed to FCN.
- Feed RNN log-signatures of different granularity at once

References I

Shujian Liao, Terry Lyons, Weixin Yang, Hao Ni
Learning stochastic differential equations using RNN with log signature features.
arXiv, 2019.

Thank you!